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seqSearch(L, x, i)
```

```
if i ≤ |L|
    if L[i] == x
        Return i
    else seqSearch(L, x, i+1)
else:
    Print("ikke fundet")
```

|L| = n

$O(n)$: linear scanning

```
BinarySearch(L, x, l, r)
```

```
if l ≤ r:
    m := ⌊(l+r)/2⌋
    if L[m] == x:
        return m
    else:
        if L[m] < x:
            return BinarySearch(L, x, m+1, r)
        else:
            return BinarySearch(L, x, l, m-1)
else:
    Print("ikke fundet")
```

$O(\log(n))$: binær scanning

$$\frac{n}{2} \quad \frac{n}{4} \quad \frac{n}{8} \quad \frac{n}{2^i} < 1 \Leftrightarrow n < 2^i \Leftrightarrow \log_2(n) < i$$

$$T(n) \in O(n)$$

$$\Leftrightarrow \exists k, n_0 : \forall n \geq n_0 : T(n) \leq f(n)$$